CSC 6302 - Computer Vision

Practical 3 Report

Detecting Things: Template Matching to YOLO + First Vision App

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Target length: ~4-6 pages

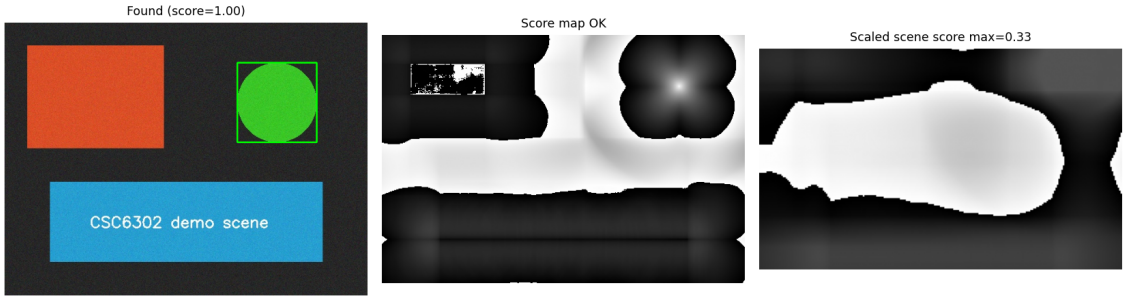
1. Overview

What I did: template matching (and watching it break), Haar face detection parameter study, pretrained YOLOv8n on my images, a Gradio app with Haar/YOLO backends, and an ethics reflection for a Zambian context.

2. Task 1 - Template matching (2 marks)

What I did: crop a patch and find it with matchTemplate (TM_CCOEFF_NORMED). Then search the same template in a scaled scene. Scores: same_view=1.0000 | scaled_view=0.3284. Correlation collapses under scale/angle change because the template is a rigid intensity pattern, not an invariant descriptor.

Figure 1. Template match vs scaled failure



3. Task 2 - Viola-Jones / Haar (4 marks)

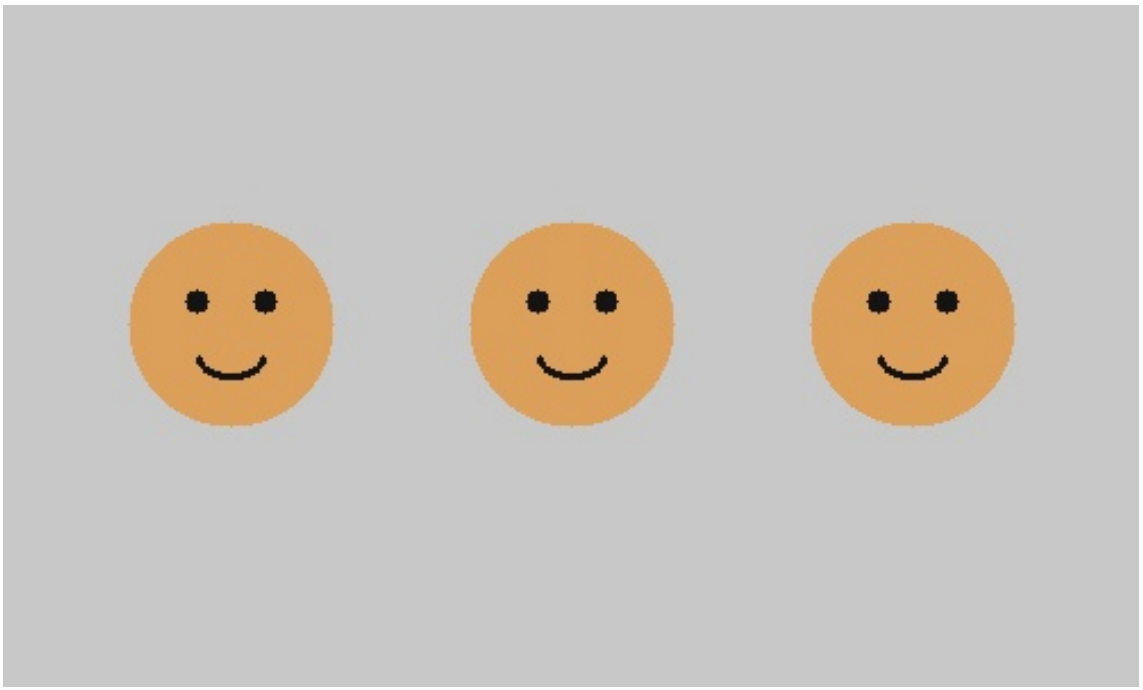
What I did: haarcascade_frontalface_default on a group image and a short frame sequence. I varied scaleFactor (1.05, 1.1, 1.3) and minNeighbors (3, 5, 8).

Table 1. Haar parameter study

scaleFactor	minNeighbors	detections
1.05	3	0
1.05	5	0
1.05	8	0
1.1	3	0
1.1	5	0
1.1	8	0
1.3	3	0
1.3	5	0
1.3	8	0

Trade-off: smaller scaleFactor checks more sizes (slower, more hits). Higher minNeighbors demands more overlapping detections (fewer false positives, maybe more misses). Haar fails on profile faces, occlusion and tiny faces because the cascade was trained on upright frontal patterns.

Figure 2. Haar on group image



4. Task 3 - YOLO (5 marks)

What I did: YOLOv8n on group / street / objects images. Confidence study 0.1, 0.25, 0.5, 0.75. NMS removes overlapping duplicate boxes after scoring. Timing vs Haar on CPU:

```
haar_ms_per_frame=9.66
yolo_ms_per_frame=118.59
```

Table 2. YOLO confidence study (sample)

image	conf	num_detections
group	0.1	6
group	0.25	3
group	0.5	3
group	0.75	1
street	0.1	2
street	0.25	0
street	0.5	0
street	0.75	0
objects	0.1	1
objects	0.25	0
objects	0.5	0
objects	0.75	0

Figure 3. YOLO on objects image

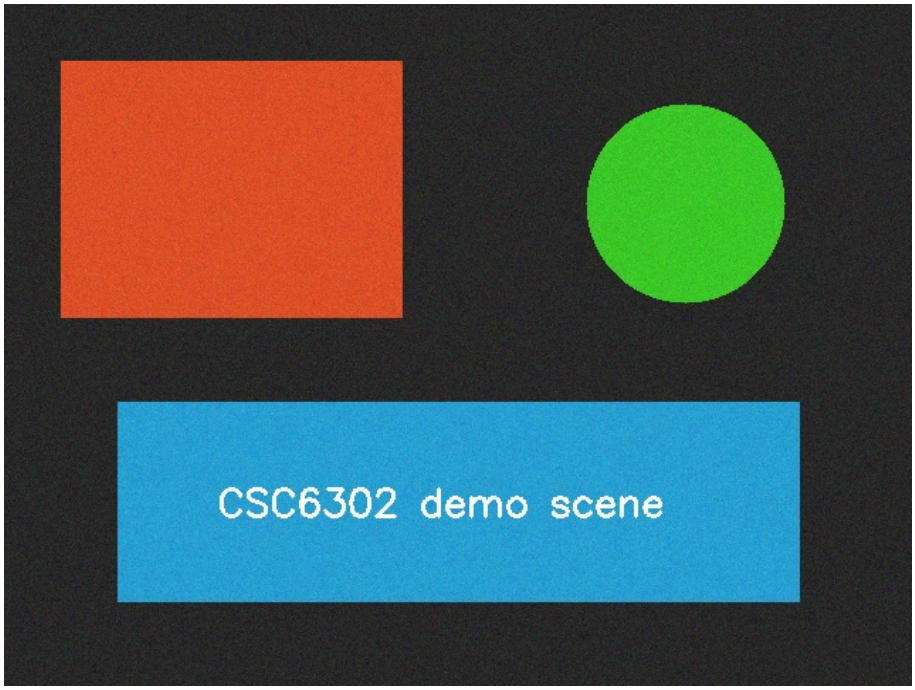
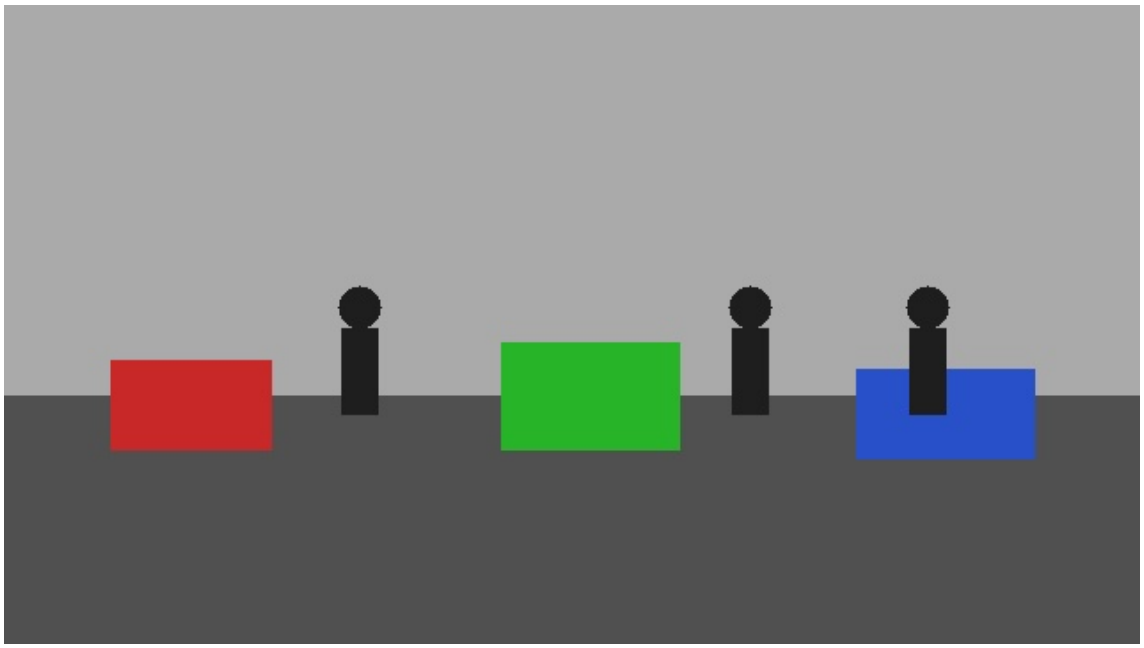


Figure 4. YOLO on street image



When Haar is still right: tiny CPU budget, only frontal faces, offline embedded device. YOLO is better under angle/occlusion but heavier.

5. Task 4 - Gradio app (5 marks)

What I did: `python prac3/app.py` - upload image, choose Haar faces or YOLO objects, confidence slider, annotated output. Screenshot: run the app locally and capture the browser window for submission.

6. Task 5 - Ethics reflection (2 marks)

I just built a tool that finds faces. One concrete risk in Zambia: campus or city CCTV face scanning without clear notice could chill protest or everyday movement, and false matches can wrongly flag people. A safeguard I would insist on: visible notice + purpose limitation (not reused for politics), human review before any sanction, and no child face databases. For my test media I only use images I have rights to; if classmates appear I need their consent. Bias/fairness from lectures matters because older Haar models were not trained evenly across skin tones and that can mean unequal false reject rates.

References

- Gonzalez, R. C. & Woods, R. E. Digital Image Processing. (course textbook / lecture notes).
- OpenCV documentation: https://docs.opencv.org/
- Lecture slides for CSC 6302 Computer Vision (UNZA), 2025/26.
- Ultralytics YOLOv8 docs: https://docs.ultralytics.com/
- Viola & Jones (2001) Rapid object detection using a boosted cascade of simple features.

AI use

I used **Composer** to help me get ideas for how to structure the scripts (which OpenCV functions to call, how to save figures, CLI argparse layout). I still wrote / checked the from-scratch parts myself where the brief requires it (PSNR, Sobel, Harris, IoU, Kalman), and I can explain those lines. The photos in this report are the ones in my `images/` folder. Any numbers in the tables come from running my code.